Report for CDA 3201L Lab #7

Sequential Circuits – Gray Code Counter

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I. Introduction

The purpose of this lab is to deepen our understanding of sequential circuit design by designing and implementing a 3-bit Gray code up counter using D flip-flops (DFFs). Unlike a standard binary counter, a Gray code counter changes only one bit per clock cycle, making it useful for minimizing switching errors in digital systems. We used the 74LS74 dual-DFF IC as our memory element, minimized the combinational next-state logic using Karnaugh maps, and implemented the resulting NAND-NAND two-level logic on a breadboard. The circuit output was displayed using three LEDs and later extended to include a 74LS47N BCD-to-7-segment decoder/driver and a 7-segment display.

II. Methods and Materials

We used the following components and equipment:

- Two SN74LS109AN dual positive edge-triggered JK flip-flop ICs (configured as DFFs by connecting $J = D$ and $\overline{K} = D$; provides both Q and $\overline{Q}$ outputs; asynchronous active-low CLR used for reset)
- One SN74LS08N quad 2-input AND gate IC
- One SN74LS86AN quad 2-input XOR gate IC
- One SN74LS04N hex inverter IC
- Three LEDs and current-limiting resistors (for original circuit)
- SN74LS47N BCD-to-7-segment decoder/driver and common-anode 7-segment display (for modified circuit)
- One push button and pull-up resistor (for active-low reset)
- Breadboard and jumper wires
- 5 V DC power supply
- Function generator (1 Hz square wave, CMOS output, for clock)

We first derived the 3-bit Gray code next-state table for the sequence $000 \to 001 \to 011 \to 010 \to 110 \to 111 \to 101 \to 100 \to 000$. Since the DFF characteristic equation is $Q^* = D$, the D inputs equal the desired next states directly. We filled in three Karnaugh maps to obtain minimized SOP expressions for D_2, D_1, and D_0. The SN74LS109AN is a JK flip-flop with a $\overline{K}$ input; connecting $J = D$ and $\overline{K} = D$ configures it as a DFF. Its $\overline{Q}$ outputs supply all complemented state signals needed by the combinational logic. The simplified equations were implemented using AND gates (SN74LS08N) and XOR gates (SN74LS86AN), with the SN74LS04N providing the single inversion required for D_0. After confirming the design with the TA, we wired the circuit on the breadboard, connected a 1 Hz clock from the function generator, and verified correct operation us-

ing the LED outputs. In the extension, we replaced the LEDs with the SN74LS47N and 7-segment display.

III. Results

1. Verilog Behavioral Model, Testbench, and Waveform

The following Verilog module describes the 3-bit Gray code counter behaviorally using a `case` statement that encodes the Gray code sequence directly:

```
module gray_counter (
    input        clk,
    input        reset,
    output reg [2:0] Q
);
    always @(posedge clk or posedge reset) begin
        if (reset)
            Q <= 3'b000;
        else begin
            case (Q)
                3'b000: Q <= 3'b001;
                3'b001: Q <= 3'b011;
                3'b011: Q <= 3'b010;
                3'b010: Q <= 3'b110;
                3'b110: Q <= 3'b111;
                3'b111: Q <= 3'b101;
                3'b101: Q <= 3'b100;
                3'b100: Q <= 3'b000;
                default: Q <= 3'b000;
            endcase
        end
    end
endmodule
```

Figure 1: *Listing 1 – Behavioral Verilog model for the 3-bit Gray code counter.*

```
module gray_counter_tb;
    reg        clk, reset;
    wire [2:0] Q;

    gray_counter uut (
        .clk   (clk),
        .reset (reset),
        .Q     (Q)
    );

    // 10 ns period clock
    initial clk = 0;
    always #5 clk = ~clk;

    initial begin
        $dumpfile("gray_counter.vcd");
        $dumpvars(0, gray_counter_tb);

        reset = 1;
        #12;          // hold reset for just over one rising edge
        reset = 0;

        #160;         // run for more than two full counter cycles
        $finish;
    end

    initial
        $monitor("%0t  clk=%b  reset=%b  Q=%b", $time, clk, reset, Q);
endmodule
```

Figure 2: *Listing 2 – Testbench for the 3-bit Gray code counter.*

The simulated waveform below shows the clock, reset, and $Q_2Q_1Q_0$ output signals cycling correctly through the Gray code sequence. After reset is de-asserted, the counter advances one state per rising clock edge, completing a full 8-state cycle before repeating.

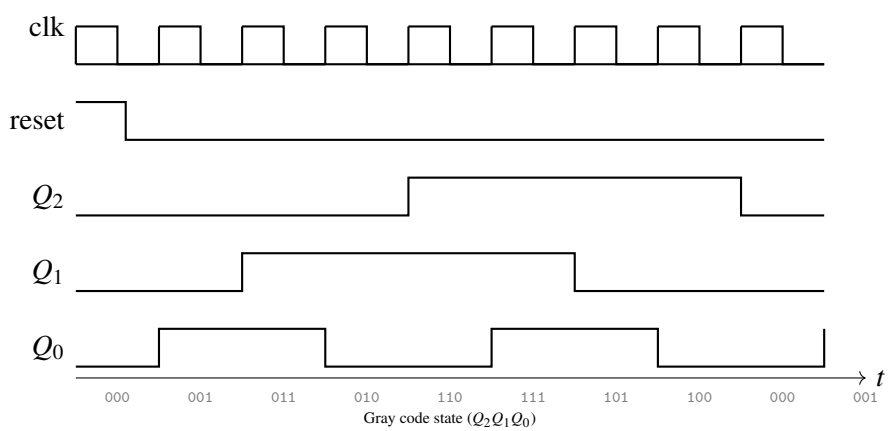

Figure 3: *Simulated waveform for the 3-bit Gray code counter. After reset, the output $Q_2Q_1Q_0$ advances through the complete Gray code sequence (000, 001, 011, 010, 110, 111, 101, 100) on successive rising clock edges. Only one bit changes at each transition, confirming correct Gray code behavior.*

2. State Transition Diagram and K-maps for D_2, D_1, D_0

Next-State Table

The 3-bit Gray code sequence defines the following next-state (and thus D-input) table:

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	D_2	D_1	D_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	1	0	1	1
0	1	1	0	1	0	0	1	0
0	1	0	1	1	0	1	1	0
1	1	0	1	1	1	1	1	1
1	1	1	1	0	1	1	0	1
1	0	1	1	0	0	1	0	0
1	0	0	0	0	0	0	0	0

State Transition Diagram

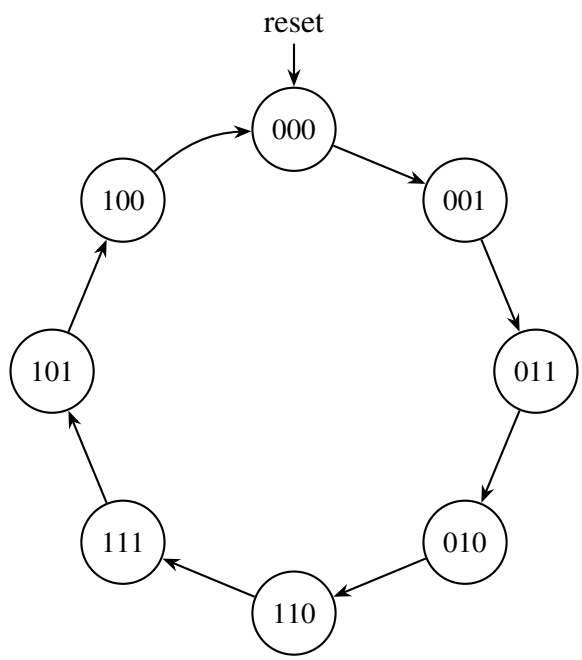

Figure 4: *State transition diagram for the 3-bit Gray code counter. Each state is labeled with its Gray code value $Q_2Q_1Q_0$. There are no external inputs; the counter advances one state per rising clock edge and cycles back to 000 after 100.*

Karnaugh Map for D_2

D_2		Q_1Q_0			
		00	01	11	10
Q_2	0	0	0	0	$\boxed{1}$
	1	0	$\boxed{1}$	$\boxed{1}$	$\boxed{1}$

- **Group 1 (2 cells):** (Q_2=0, Q_1Q_0=10) and (Q_2=1, Q_1Q_0=10): $Q_1 = 1$, $Q_0 = 0 \Rightarrow Q_1\overline{Q_0}$.
- **Group 2 (2 cells):** (Q_2=1, Q_1Q_0=01) and (Q_2=1, Q_1Q_0=11): $Q_2 = 1$, $Q_0 = 1 \Rightarrow Q_2Q_0$.

$$\boxed{D_2 = Q_1\overline{Q_0} + Q_2Q_0}$$

Karnaugh Map for D_1

D_1		Q_1Q_0			
		00	01	11	10
Q_2	0	0	$\boxed{1}$	$\boxed{1}$	$\boxed{1}$
	1	0	0	0	$\boxed{1}$

- **Group 1 (2 cells):** (Q_2=0, Q_1Q_0=01) and (Q_2=0, Q_1Q_0=11): $Q_2 = 0$, $Q_0 = 1 \Rightarrow \overline{Q_2}Q_0$.
- **Group 2 (2 cells):** (Q_2=0, Q_1Q_0=10) and (Q_2=1, Q_1Q_0=10): $Q_1 = 1$, $Q_0 = 0 \Rightarrow Q_1\overline{Q_0}$.

$$\boxed{D_1 = \overline{Q_2}Q_0 + Q_1\overline{Q_0}}$$

Note that the term $Q_1\overline{Q_0}$ appears in both D_2 and D_1, so the corresponding NAND gate is shared in the hardware implementation.

Karnaugh Map for D_0

D_0		Q_1Q_0			
		00	01	11	10
Q_2	0	$\boxed{1}$	$\boxed{1}$	0	0
	1	0	0	$\boxed{1}$	$\boxed{1}$

- **Group 1 (2 cells):** (Q_2=0, Q_1Q_0=00) and (Q_2=0, Q_1Q_0=01): $Q_2 = 0$, $Q_1 = 0 \Rightarrow \overline{Q_2}\,\overline{Q_1}$.
- **Group 2 (2 cells):** (Q_2=1, Q_1Q_0=11) and (Q_2=1, Q_1Q_0=10): $Q_2 = 1$, $Q_1 = 1 \Rightarrow Q_2Q_1$.

$$\boxed{D_0 = \overline{Q_2}\,\overline{Q_1} + Q_2Q_1}$$

This expression is equivalent to $\overline{Q_2 \oplus Q_1}$ (XNOR), meaning D_0 is 1 whenever Q_2 and Q_1 are equal.

AND-XOR-INV Implementation

Observing that the product terms in D_2 and D_1 are mutually exclusive (they cannot both be 1 at the same time), the OR can be replaced by XOR:

$$\begin{aligned}
D_2 &= (Q_1 \cdot \overline{Q_0}) \oplus (Q_2 \cdot Q_0) \\
D_1 &= (\overline{Q_2} \cdot Q_0) \oplus (Q_1 \cdot \overline{Q_0}) \\
D_0 &= \overline{Q_2 \oplus Q_1}
\end{aligned}$$

This maps directly onto the available ICs:

- **SN74LS08N (AND):** compute $Q_1 \cdot \overline{Q_0}$ (shared between D_2 and D_1), $Q_2 \cdot Q_0$, and $\overline{Q_2} \cdot Q_0$ — 3 of 4 gates used.
- **SN74LS86AN (XOR):** compute D_2, D_1, and $Q_2 \oplus Q_1$ — 3 of 4 gates used.
- **SN74LS04N (INV):** invert $Q_2 \oplus Q_1$ to produce D_0 — 1 of 6 inverters used.

All complemented state signals ($\overline{Q_0}$, $\overline{Q_1}$, $\overline{Q_2}$) are taken directly from the $\overline{Q}$ outputs of the SN74LS109AN flip-flops, requiring no additional inverters for those signals. The term $Q_1 \cdot \overline{Q_0}$ is shared between D_2 and D_1, reducing the AND gate count to three.

3. Wiring Diagrams

Original Circuit – Gray Code Counter with LED Output

The circuit uses two SN74LS109AN ICs (three of the four JK flip-flops are used for Q_2, Q_1, Q_0), one SN74LS08N (AND), one SN74LS86AN (XOR), and one SN74LS04N (inverter). Each flip-flop is configured as a DFF by connecting both J and $\overline{K}$ to the same D input signal. The key wiring connections are:

Signal	Connection
CLK (all 3 FFs)	Function generator 1 Hz CMOS output
$\overline{\text{CLR}}$ (all 3 FFs)	Reset button (active-low; pull-up resistor to V_{DD})
$\overline{\text{PRE}}$ (all 3 FFs)	V_{DD} (preset disabled)
$J_2 = \overline{K}_2 = D_2$	XOR output: $(Q_1 \cdot \overline{Q_0}) \oplus (Q_2 \cdot Q_0)$
$J_1 = \overline{K}_1 = D_1$	XOR output: $(\overline{Q_2} \cdot Q_0) \oplus (Q_1 \cdot \overline{Q_0})$
$J_0 = \overline{K}_0 = D_0$	INV output: $\overline{Q_2 \oplus Q_1}$
$\overline{Q_0}, \overline{Q_1}, \overline{Q_2}$	Taken from $\overline{Q}$ pins of SN74LS109AN (no extra inverter)
Q_2, Q_1, Q_0	LED anodes (through 330 Ω resistors to GND)

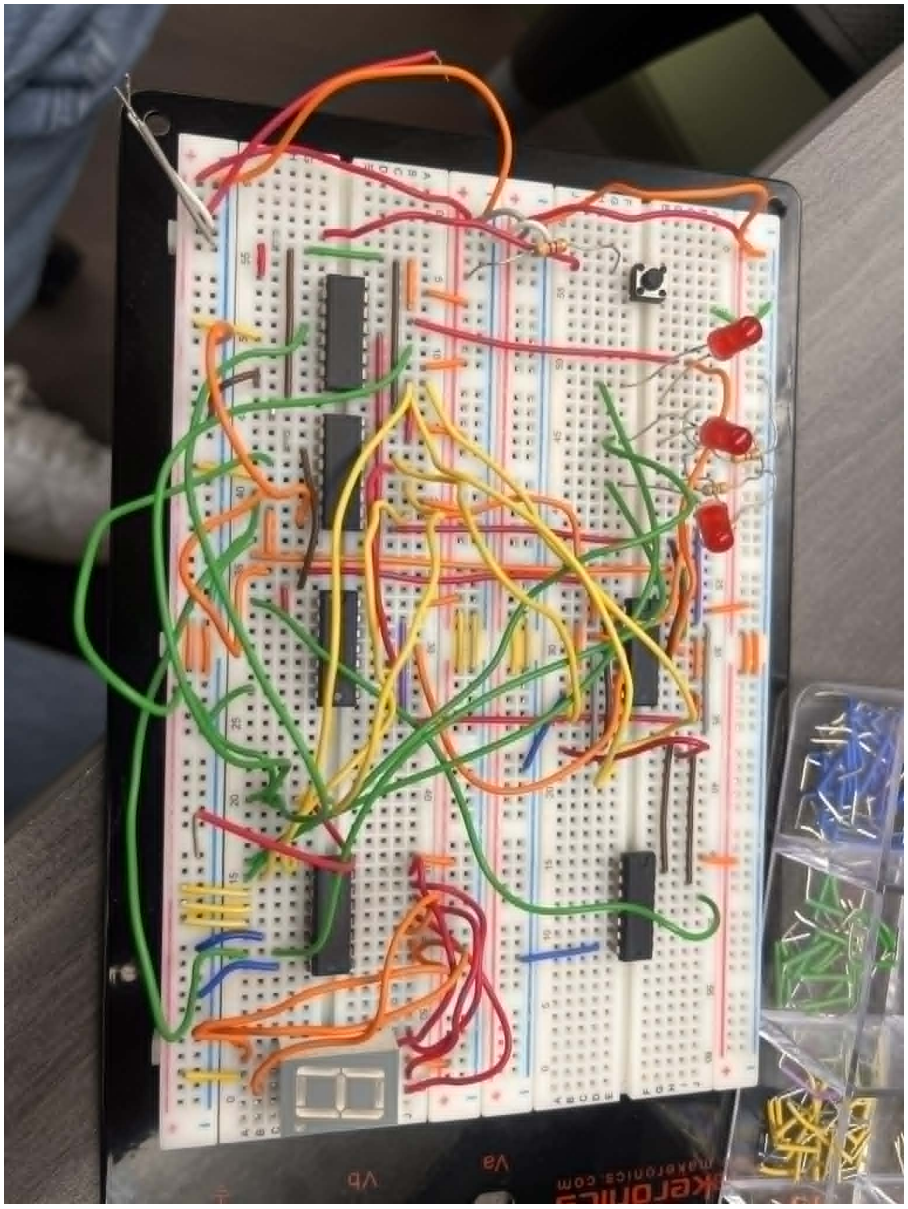

Figure 5: *Breadboard implementation of the 3-bit Gray code counter. The two 74LS74 DFF ICs (center) provide the three flip-flops and their complemented outputs. The 74LS00 NAND IC (lower center) implements the minimized next-state logic. The three red LEDs (right) display the current Gray code value $Q_2Q_1Q_0$. A push button (top) is wired with a pull-up resistor to the active-low CLR pins for reset.*

Supplementary Video: Circuit Demonstration Recording (Google Drive)

Modified Circuit – Gray Code Counter with 7-Segment Display

In the modified circuit the three LEDs are replaced by the 74LS47N BCD-to-7-segment decoder/driver and a common-anode 7-segment display. The counter outputs Q_2, Q_1, Q_0 are connected to inputs C, B, A of the 74LS47N respectively (MSB to LSB). All 74LS47N segment outputs ($\overline{a}$–$\overline{g}$, active-low) drive the corresponding segments of the display through current-limiting resistors (330 Ω recommended to avoid exceeding the display's forward current rating). The $\overline{\text{LT}}$, $\overline{\text{RBI}}$, and $\overline{\text{BI/RBO}}$

inputs on the 74LS47N are tied to V_{DD} to keep lamp-test and blanking inactive. The rest of the counter circuit (DFFs, NAND logic, clock, reset) remains unchanged.

74LS47N Pin	**Connection**
A (pin 7)	Counter Q_0
B (pin 1)	Counter Q_1
C (pin 2)	Counter Q_2
D (pin 6)	GND (no 4th bit)
$\overline{\text{LT}}$ (pin 3)	V_{DD}
$\overline{\text{RBI}}$ (pin 5)	V_{DD}
$\overline{\text{BI/RBO}}$ (pin 4)	V_{DD}
Segment outputs $\overline{a}$–$\overline{g}$ (pins 9–15)	7-segment display segments a–g via 330 Ω
Common anode (display)	V_{DD}
V_{CC} (pin 16)	5 V
GND (pin 8)	GND

IV. Discussion

The circuit behaved as expected. After asserting reset, all three LEDs were off ($Q_2Q_1Q_0 = 000$). With the function generator providing a 1 Hz clock, the LEDs cycled through the Gray code sequence at a rate slow enough to observe each state change visually. Crucially, only one LED changed state at each clock edge, confirming the single-bit-change property of Gray code and validating our next-state logic.

The K-map simplification produced compact excitation equations. Because the two product terms in D_2 and D_1 are mutually exclusive, OR can be replaced by XOR, enabling a clean implementation with SN74LS08N AND gates and SN74LS86AN XOR gates. The term $Q_1 \cdot \overline{Q_0}$ is shared between D_2 and D_1, reducing the AND gate count to three (within a single SN74LS08N IC). The SN74LS109AN JK flip-flop, configured as a DFF by connecting $J = \overline{K} = D$, provided both Q and $\overline{Q}$ outputs natively, supplying all complemented state signals without requiring additional inverter gates. The SN74LS04N was used solely to invert the XOR output for $D_0 = \overline{Q_2 \oplus Q_1}$.

The expression $D_0 = \overline{Q_2 \oplus Q_1}$ has an intuitive interpretation: the LSB of the next Gray code state is 1 whenever the two MSBs of the current state are equal, and 0 otherwise. This reflects the mirror symmetry of the Gray code sequence across its midpoint.

In the 7-segment display extension, the display shows digit values 0–7 in the order 0, 1, 3, 2, 6, 7, 5, 4 (the decimal equivalents of the Gray code words), which is visually non-monotonic but demonstrates the counter's correct Gray-code sequencing.

V. Conclusion

In this lab we designed and implemented a 3-bit Gray code up counter. We derived the next-state table, simplified the excitation equations for D_2, D_1, and D_0 using Karnaugh maps, and implemented the logic using SN74LS08N AND gates and SN74LS86AN XOR gates, with a single SN74LS04N inverter for D_0. The SN74LS109AN JK flip-flops, configured as DFFs via $J = \overline{K} = D$, served as the memory elements and provided $\overline{Q}$ outputs that eliminated the need for extra inverters. A Verilog behavioral model and testbench confirmed the expected waveform. The breadboard circuit, clocked at 1 Hz, successfully cycled through all eight Gray code states with exactly one

LED changing per clock edge. The optional extension connected the counter to a SN74LS47N and 7-segment display, further demonstrating the circuit's correct operation.